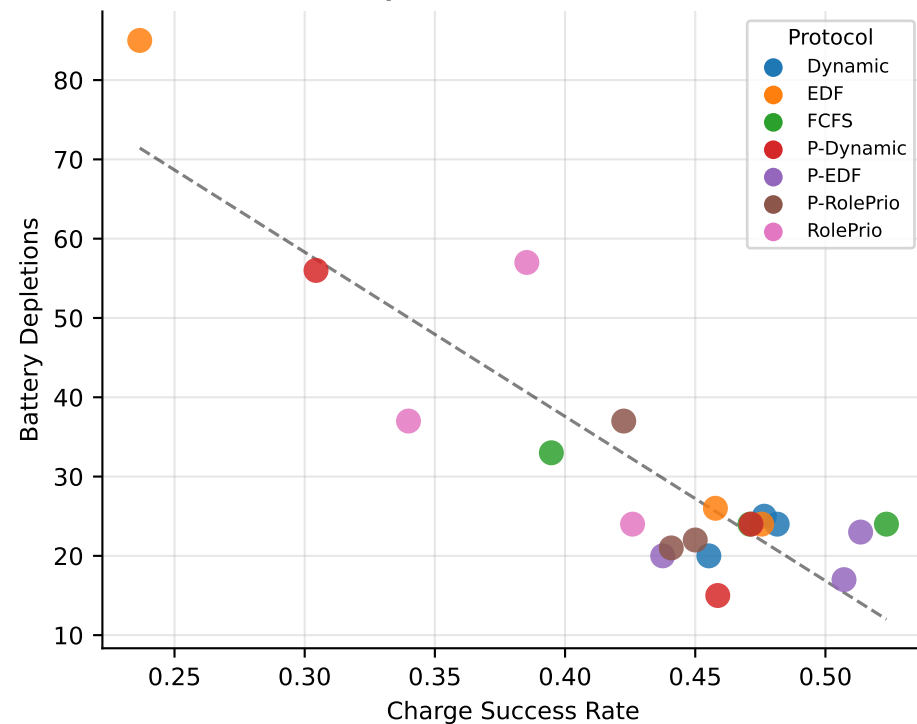
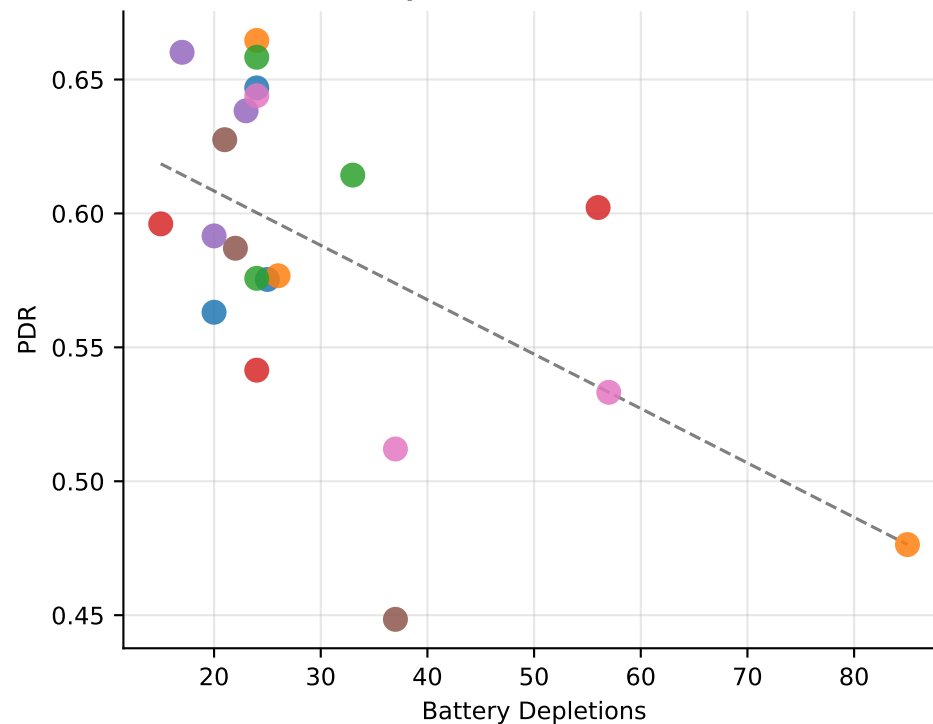


Cross-Layer Mechanism: Charging → Survivability → Network (Three-step causal chain; n = 21 runs)

Charging → Fleet Survival
 $r = -0.87$, $\rho = -0.56$
 $p = 0.0000, 0.0078$



Fleet Survival → Network PDR
 $r = -0.57$, $\rho = -0.48$
 $p = 0.0074, 0.0290$



Routing Drops → Network PDR
 $r = -0.53$, $\rho = -0.55$
 $p = 0.0140, 0.0091$

